

Deliverable 1

Vision Assist Innovators
Smart Assistive System for Visually Impaired People
GROUP 5

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Table of Contents

1. Document Title - Page 1
2. Team Name - Page 1
3. Project Name - Page 1
4. Team Number - Page 1
5. Student's names and IDs - Page 1
6. Table of Contents - Page 1
7. Team Contract (from CENG-322_TEAM_contract_v2.0) - Page 2
8. Signature of each student who participated in the work (team members who did not participate and where their signature is missing will get 0) - Page 3
9. In the pdf, write the requirement and comment on it for each item. Include the question and the answer - Page 4
10. Create Google Play Developer Account, take a screenshot showing account has been approved and created (one or more members) - Page 5
11. Link GitHub Repo. All members must contribute to the repo - Page 5
12. Each member must have a minimum of three commits. Missing commits will get 0. Commits must be done from Android Studio - Page 5
13. Screenshot of Github invitation - Page 6
14. Project Background and Description
 1. Describe the project goals and final vision - Page 6
 2. Describe the software aspect and hardware - Page 6
 3. Describe the screen flows - Page 6
 4. How you incorporated the feedback provided through the interview - Page 7
 5. Demonstrate how you are planning to satisfy to read / write from the DB which is hosted on the cloud - Page 7
15. Project Scope: Describe the technical scope of the project by talking about the project plan, and how you will know when the project is complete - Page 8
16. Describe how to integrate the software with the hardware components - Page 9
17. Project layout: write how you plan to implement Navigation Drawer, and combination with another layout - Page 9
18. Write Minimum of 1 theme: for the Theme write: 1. Two epics, 3 stories for each epic, 3 tasks for each story - Page 11
19. If document not in pdf, or not the proper name (marks will be deducted!) - Page 12
20. Please save the PDF document, i.e. TeamName_ProjectName_GroupNumber_Deliverable_1.pdf, and submit on blackboard - Page 12
21. The last submitted pdf file will be marked - Page 12

CENG-322 TEAM PROJECT

Team Name: __Vision Assist Innovators__

Team Number: ____05_____

Project Name: __Smart Assistive System _____

Please negotiate, sign, scan and include as the first section in your Deliverable 1.

Please note that if cheating is discovered in a group assignment each member will be charged with a cheating offense regardless of their involvement in the offense. Each member will receive the appropriate sanction based on their individual academic honesty history.

Please ensure that you understand the importance of academic honesty. Each member of the group is responsible to ensure the academic integrity of all of the submitted work, not just their own part. Placing your name on a submission indicates that you take responsibility for its content.

Team Member Names (Please Print)	Signatures	Student ID	Github Id
Project Leader: • Sarang Prajapati	Sarang	– N01662036	SarangParjapati2036
 • Krish Patel	Krish	– N01666556	KrishPatel6556

• Kush Patel	Kush	– N01657387	KushPatel7387
• Daksh Rana – N01664095	Rana	– N01664095	dakshrana4095

For further information read Academic Honesty Policy on <https://humber.ca/legal-and-risk-management/policies/search-by-students.html>.

By signing this contract, we acknowledge having read the Humber Academic Honesty Policy as per the link below.

<https://academic-regulations.humber.ca/2018-2019/17.0-ACADEMIC-MISCONDUCT>

Responsibilities of the Project Leader include:

- Assigning tasks to other team members, including self, in a fair and equitable manner.
- Ensuring work is completed with accuracy, completeness and timeliness.
- Planning for task completion to ensure timelines are met
- Any other duties as deemed necessary for project completion

What we will do if . . .

Scenario	Accepted initials	We agree to do the following
Team member does not deliver component on time due to severe illness or extreme personal problem	A	a) Team absorbs workload temporarily ____ d) Other:
Team member cannot deliver component on time due to lack of ability	B	a) Team reassigns component ____ b) Team helps member ____

Scenario	Accepted initials	We agree to do the following
		c) 'Team "fires" team member by not permitting his/her name on submission ____ d) Other:
Team member does not deliver component on time due to lack of effort	B	a) 'Team absorbs workload ____ b) 'Team "fires" team member by not permitting his/her name on submission ____ c) Other:
Team member does not attend team meeting	A	a) 'Team proceeds without him/her and will assign work to the absent member ____ b) 'Team doesn't proceed and records team member's absence ____ c) 'Team proceeds for that meeting but "fires" member after ____ occurrences ____
An unforeseen constraint occurs after the deliverable has been allocated and	A	a) 'Team meets and reschedules deliverable ____

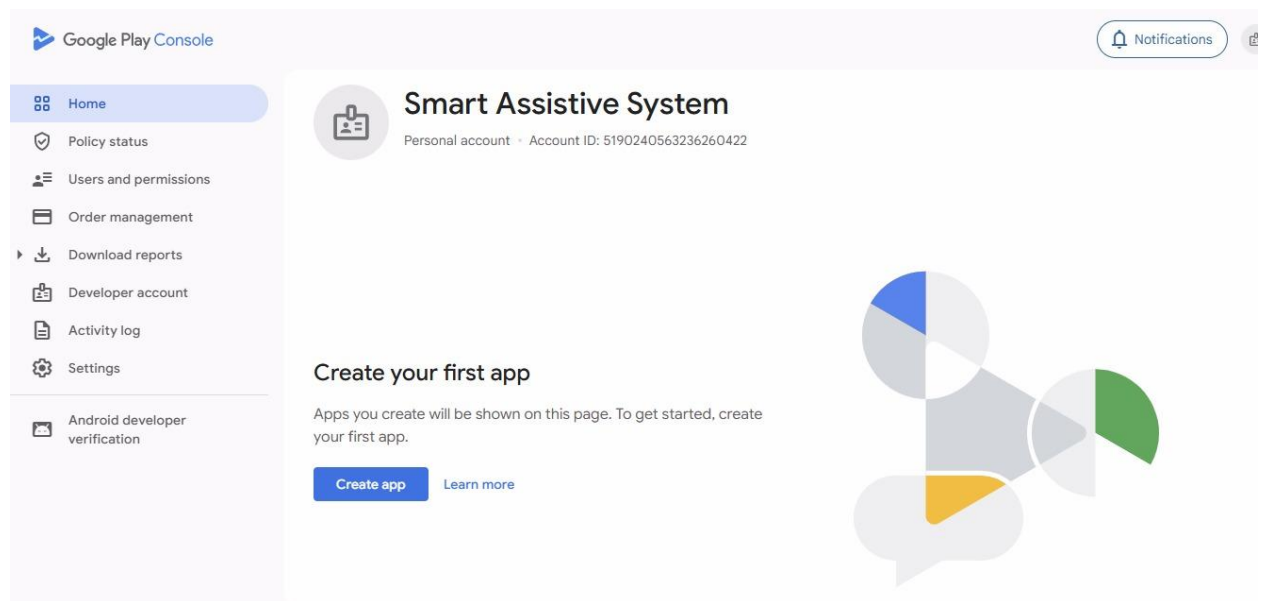
Scenario	Accepted initials	We agree to do the following
scheduled (a surprise test or assignment)		b) Team will cope with constraint ____ c) Other:
Team cannot achieve consensus leaving one member feeling "railroaded", "ignored", or "frustrated" with a decision which affects all parties	A	a) Team agrees to abide by majority vote ____ b) Team flips coin ____ c) Other:
Team members do not share expectations for grade desired	A	a) Team will elect one person as "standards-bearer" who has the right to ask that work be redone ____ b) Team votes on each submission's quality ____ c) Team will ask for individual marking and will identify sections by author ____ d) Other:
Team member behaves in an unprofessional manner by being rude or uncooperative	A	a) Team attempts to resolve the issue by airing the problem at team meeting ____ b) Team ignores behaviour ____

Scenario	Accepted initials	We agree to do the following
		c) Team agrees to avoid use of all vocabulary inappropriate to the business setting ____ d) Team fires the team member.
Team member assumes or requests that his/her name be signed to a submission but has not participated in production of the deliverable	A	a) Team agrees that this is cheating and is unethical ____ b) Friends are friends and should help each other ____ c) That person name will not be put on the submission._
There is a dominant team member who is content to make all decisions on the team's behalf leaving some team members feeling like subordinates rather than equal members	A	a) Team will actively solicit consensus on all decisions which affect project direction by asking for each member's decision and vote ____ b) Team will express subordination feelings and attempt to resolve issue ____ c) Other:
Team has a member who refuses to participate in decision making but complains to others that s/he wasn't consulted	A	a) Team forces decision sharing by routinely voting on all issues ____

Scenario	Accepted initials	We agree to do the following
		b) Team routinely checks with each other about perceived roles ____ c) Team discusses the matter at team meeting ____

9)???????

10) Google Play Developer Account



11) GIT REPOSITORY LINK

GitHub



@PatelKush7387 has invited you to collaborate on the
[PatelKush7387/SmartAssistiveSystem](#) repository

You can [accept or decline](#) this invitation. You can also visit [@PatelKush7387](#) to learn a bit more about them.

This invitation will expire in 7 days.

[View invitation](#)

13) Git Hub Invitation Link and screenshot:-

<https://github.com/KushPatel7387/SmartAssistiveSystem/invitations>

14) Project Background and Description:

i) Our Project goal is to have blind people safely navigate with a smart system. Our goal is to use sensors and an app to give real-time alerts. The final vision is a portable device that warns about obstacles, colors, and light, making life easier and safer

ii) We are going to use TSL2591 (light sensor), TCS34725 (color sensor), VL53L1X (distance sensor), and AS7262/AS7341 (spectrum sensor) with Raspberry pi as a controller. Moreover, for the software, we will prefer to use Java as the main Language and for application making process.

lii) The app starts with a 3-second splash screen, then goes to a main screen with a nav drawer. Options include a dashboard (live sensor

data), settings, and emergency. Tabs show sensor details, and bottom nav switches fast.

iv) In the interview , we guys were suggested to embed GPS tracking as well as emergency call buttons with an help alert for the caretakers.

v) We'll use Firebase. The app writes sensor data like distance or color to Firebase with setValue(). It reads live data for the caretaker dashboard using addValueEventListener.

15) Project Scope:

i) The project involves developing a Smart Assistive System that integrates four sensors (light, color, distance, and spectral) with an Android app to provide real-time navigation support for visually impaired people. The plan includes hardware integration, mobile app development with Navigation Drawer and Fragments, and Bluetooth/Wi-Fi communication for live data and alerts. The project will be complete when all sensors work accurately, the app displays real-time data with instant alerts (voice, vibration, sound), supports English and French, and runs without errors or crashes.

16) Integration of Software with Hardware Components:

i) The integration of the system is achieved using an ESP32 microcontroller (or Raspberry Pi) as the central hardware interface. Each sensor—TSL2591 (light), TCS34725 (color), VL53L1X (distance), and AS7262/AS7341 (spectral)—is connected to the microcontroller through I²C or SPI communication protocols. The microcontroller continuously collects raw sensor data and performs preprocessing when required.

To connect with the Android application, the microcontroller uses Bluetooth (BLE) or Wi-Fi. The sensor readings are formatted into structured data packets (e.g., JSON) and transmitted to the mobile app. On the software side, the Android app (developed in Java/Kotlin) receives and parses the data, displaying it in designated fragments such as *Home*, *Sensors*, *Alerts*, and *About*.

The application processes the data further by generating real-time alerts—through voice, vibration, or sound—based on predefined thresholds for brightness, obstacle distance, color detection, or spectral signals. Moreover, a caretaker dashboard in the app provides live visualization of sensor data for monitoring user safety.

This two-way integration ensures that the hardware delivers accurate environmental inputs, while the software translates them into actionable and user-friendly feedback, enabling safe and independent navigation for visually impaired individuals.

17) Project Layout:

i) To implement the Navigation Drawer, our project uses the Android Navigation Component combined with a DrawerLayout. The Navigation Drawer allows users to switch between different fragments (*Home, Sensors, Alerts, About*) with a consistent and accessible interface. It is integrated with a Toolbar (AppBar) for seamless navigation.

ii) We also plan to combine the Navigation Drawer with another layout such as Bottom Navigation or Tabs depending on the section:

- 1) Navigation Drawer → for accessing high-level sections (Home, Sensors, Alerts, About).
- 2) Tabs → for sub-sections inside a fragment (e.g., switching between “Live Data” and “History” in the Sensors screen).
- 3) Bottom Navigation (alternative) → could be used for quick access to the most commonly used 3 screens.

Why Use Them:

- 1) Navigation Drawer: Best for apps with 4+ primary destinations. It keeps the main UI clean and accessible.
- 2) Bottom Navigation: Best when only 3–5 primary destinations are needed frequently. Quick access at the bottom.
- 3) Tabs: Ideal for dividing related content within the same screen (e.g., categories, lists, dashboards).

Use Cases in Existing Apps:

- 1) Navigation Drawer: Gmail (switching between Inbox, Sent, Drafts, Settings).
- 2) Bottom Navigation: Instagram (Home, Search, Reels, Shop, Profile).
- 3) Tabs: YouTube (Home, Shorts, Subscriptions, Library).

One vs. the Others

- 1) Drawer vs. Bottom Navigation: Drawer is hidden and good for less-used destinations, while Bottom Navigation is always visible for frequent actions.
- 2) Tabs vs. Drawer/Bottom Navigation: Tabs are for organizing related content within one section, whereas Drawer/Bottom Navigation manage overall app destinations.

18) Epic 1: Sensor Data Integration with Mobile App

Story 1.1: As a user, I want the app to receive real-time sensor data from hardware.

- 1) Task 1.1.1: Set up Bluetooth/Wi-Fi communication between ESP32 and Android app.
- 2) Task 1.1.2: Parse JSON packets in the Android app.
- 3) Task 1.1.3: Display live sensor values in the Sensors Fragment.

Story 1.2: As a user, I want alerts when obstacles or signals are detected.

- 1) Task 1.2.1: Implement distance threshold alerts (vibration/voice).
- 2) Task 1.2.2: Integrate TCS34725/AS7262 detection for color/traffic signals.
- 3) Task 1.2.3: Link alert outputs to Android notification system.

Story 1.3: As a caretaker, I want to view sensor data remotely.

- 1) Task 1.3.1: Create caretaker dashboard in the app.
- 2) Task 1.3.2: Fetch sensor data from ESP32 and sync with app.
- 3) Task 1.3.3: Implement live updates (e.g., every 2 seconds).

Epic 2: Android App User Experience (UX) & Navigation

Story 2.1: As a user, I want simple navigation between screens.

- 1) Task 2.1.1: Implement Navigation Drawer with fragments (Home, Sensors, Alerts, About).
- 2) Task 2.1.2: Add icons and labels for clarity.
- 3) Task 2.1.3: Connect fragments to navigation graph.

Story 2.2: As a user, I want alerts presented in multiple formats.

- 1) Task 2.2.1: Enable vibration patterns for different alerts.
- 2) Task 2.2.2: Add text-to-speech for voice guidance.
- 3) Task 2.2.3: Configure app sounds (beep tones).

Story 2.3: As a user, I want the app to support multilingual use (English & French).

- 1) Task 2.3.1: Create strings.xml (English) and strings-fr.xml (French).
- 2) Task 2.3.2: Add app settings for language preference.
- 3) Task 2.3.3: Test switching between languages.

